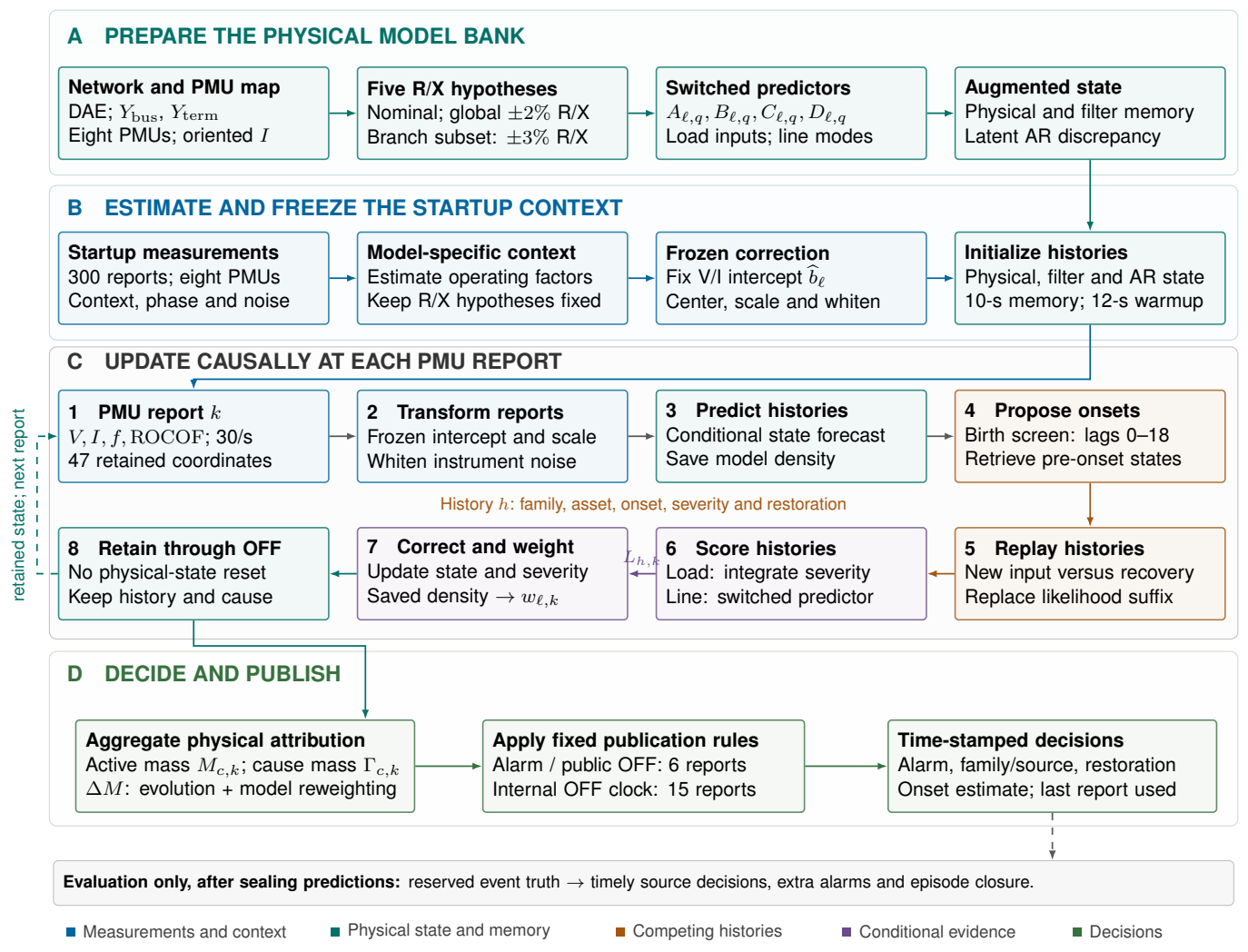

A PREPARE THE PHYSICAL MODEL BANK
Network and PMU map
DAE; Y_{bus}, Y_{term}
Eight PMUs; oriented I
Five R/X hypotheses
Nominal; global $\pm 2\%$ R/X
Branch subset: $\pm 3\%$ R/X
Switched predictors
$A_{\ell,q}, B_{\ell,q}, C_{\ell,q}, D_{\ell,q}$
Load inputs; line modes
Augmented state
Physical and filter memory
Latent AR discrepancy
B ESTIMATE AND FREEZE THE STARTUP CONTEXT
Startup measurements
300 reports; eight PMUs
Context, phase and noise
Model-specific context
Estimate operating factors
Keep R/X hypotheses fixed
Frozen correction
Fix V/I intercept $\widehat{b}_\ell$
Center, scale and whiten
Initialize histories
Physical, filter and AR state
10-s memory; 12-s warmup
C UPDATE CAUSALLY AT EACH PMU REPORT
retained state; next report
1 PMU report k
V, I, f, ROCOF; 30/s
47 retained coordinates
2 Transform reports
Frozen intercept and scale
Whiten instrument noise
3 Predict histories
Conditional state forecast
Save model density
4 Propose onsets
Birth screen: lags 0–18
Retrieve pre-onset states
History h: family, asset, onset, severity and restoration
8 Retain through OFF
No physical-state reset
Keep history and cause
7 Correct and weight
Update state and severity
Saved density $\rightarrow w_{\ell,k}$
$L_{h,k}$
6 Score histories
Load: integrate severity
Line: switched predictor
5 Replay histories
New input versus recovery
Replace likelihood suffix
D DECIDE AND PUBLISH
Aggregate physical attribution
Active mass $M_{c,k}$; cause mass $\Gamma_{c,k}$
ΔM: evolution + model reweighting
Apply fixed publication rules
Alarm / public OFF: 6 reports
Internal OFF clock: 15 reports
Time-stamped decisions
Alarm, family/source, restoration
Onset estimate; last report used
Evaluation only, after sealing predictions: reserved event truth → timely source decisions, extra alarms and episode closure.
Measurements and context
Physical state and memory
Competing histories
Conditional evidence
Decisions